

Swetha Manchukonda

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Professional Summary

Data Analyst and BI Developer with 3 years of experience building Power BI dashboards, writing advanced SQL queries, and developing Python-based data processing pipelines across enterprise environments. Proven ability to design scalable data models, optimize query performance, and deliver actionable business insights to cross-functional stakeholders. Hands-on experience with Snowflake, Azure Synapse Analytics, and cloud-based data platforms. Familiar with AI productivity tools and Microsoft Copilot. Master's in Data Science, GPA: 3.87/4.0.

Technical Skills

- **BI & Visualization:** Power BI (KPI Dashboards, DAX, Executive Reporting), Tableau, Streamlit, Matplotlib, Seaborn
- **SQL & Databases:** Advanced SQL (CTEs, Window Functions, Multi-Table JOINS, Subqueries), Snowflake, SQL Server, Oracle, Query Optimization
- **Python & Automation:** Pandas, NumPy, Matplotlib, Seaborn, Data Processing Automation, ETL Scripting, Flask
- **Data Modeling & Warehousing:** Snowflake, Azure Synapse Analytics, Azure Data Factory, PySpark, Databricks, GCP
- **AI & Productivity Tools:** Microsoft Copilot, GitHub Copilot, Jupyter Notebook, VS Code, Git
- **Machine Learning:** Scikit-learn, Random Forest, XGBoost, Logistic Regression, Feature Engineering, SMOTE
- **Data Quality:** Null Analysis, Duplication Checks, Distribution Validation, UAT Execution, SOP Documentation

Professional Experience

Research Assistant - Data Analyst

Aug 2025 – Present

University of Maryland Baltimore County (UMBC)

- Developed and maintained Power BI dashboards and reports delivering real-time KPI tracking and operational performance metrics to senior leadership and cross-functional stakeholders.
- Wrote highly complex SQL queries utilizing CTEs, window functions, multi-table JOINS, and nested subqueries — reducing data processing latency by 20% through query optimization and pipeline restructuring.
- Built and managed data models integrating multiple source systems using Python, SQL, PySpark, and Azure Data Factory, enabling predictive modeling and ad hoc business analysis at enterprise scale.
- Worked with Snowflake and Azure Synapse Analytics to develop recurring extracts and analytic views, ensuring data accuracy and performance optimization across high-volume datasets.
- Ensured data accuracy through null analysis, duplication checks, and distribution validation — achieving 99% data accuracy across high-volume datasets.
- Leveraged Microsoft Copilot and AI-assisted tools to improve development efficiency and accelerate dashboard delivery timelines.
- Translated ambiguous stakeholder requirements into structured SQL queries, scalable analytical workflows, and actionable business insights.

Data Analyst

Jul 2022 – Jul 2023

Cognizant Technology Solutions Pvt Ltd

- Developed Power BI dashboards and recurring data extracts for a global enterprise client, improving reporting accuracy and enabling faster operational decisions across cross-functional teams.
- Wrote and optimized SQL queries for data extraction, transformation, and reporting across relational databases — improving query performance and reducing report generation time.
- Identified and resolved critical data quality issues during UAT cycles through null analysis, duplication checks, and distribution validation — preventing downstream reporting errors.
- Built Python-based automation scripts to streamline data migration workflows and custom reporting pipelines, reducing manual effort across stakeholder teams.

- Maintained SOP-style data dictionaries and process documentation to support data governance, audit readiness, and consistent reporting standards across global teams.

Data Analyst Trainee

Sep 2021 – Jun 2022

Cognizant Technology Solutions Pvt Ltd

- Developed proficiency in advanced SQL querying, relational database design, data modeling, and ETL workflows through structured training aligned with real-world business analytics use cases.
- Practiced data extraction, transformation, and validation workflows including null checks and duplication analysis across simulated analytics projects.
- Built data-driven web interfaces using JavaScript to present analytical outputs, improving how business users consumed and acted on structured data insights.

Key Projects

Fake Job Post Detection | *Python, Scikit-learn, TF-IDF, Random Forest, PySpark*

- Built a machine learning classification pipeline benchmarking Random Forest, Logistic Regression, Naive Bayes, and XGBoost — achieving 95%+ accuracy through TF-IDF feature engineering and rigorous data validation.
- Deployed solution as a Streamlit web application enabling real-time risk scoring for non-technical end users, with full SOP-style documentation for all data files and feature sets.

Loan Approval Risk Scoring | *Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn*

- Developed a member-level risk scoring model improving loan approval decision accuracy by 40% through advanced feature engineering from raw structured financial data.
- Resolved a 40% class imbalance using SMOTE and delivered executive report with model performance visualizations translating findings into actionable recommendations for senior decision-makers.

Education

Master of Professional Studies in Data Science

Aug 2023 – May 2025

University of Maryland Baltimore County

GPA: 3.87/4.0

B.Tech, Electronics & Communication Engineering

Aug 2017 – Jul 2021

CVR College of Engineering, Hyderabad, India